DRUGS · ALLERGY, RHINITIS AND BIOLOGICS

Montelukast

Montelukast blocks the cysteinyl leukotriene receptor. It is a modest treatment for allergic rhinitis, best suited to patients who also have asthma, and it has a role in aspirin-exacerbated respiratory disease and, as an alternative to surgery, in children with mild obstructive sleep-disordered breathing. It is a once-daily tablet with an important warning: neuropsychiatric reactions, including in children.

1 CHAPTER ONE At a glance

The drug on one page.

1.1 Summary

MONTELUKAST

	SUMMARY
Class	Leukotriene receptor antagonist
How it works	Blocks the CysLT1 receptor, reducing bronchoconstriction, mucosal oedema, mucus and eosinophil recruitment
ENT uses	Allergic rhinitis with asthma; aspirin-exacerbated respiratory disease (Samter's triad) with nasal polyps; mild paediatric obstructive sleep apnoea (with nasal steroids) as an alternative or bridge to adenotonsillectomy
Adult dose	10 mg once daily in the evening
Child dose	6 months to 5 years 4 mg; 6 to 14 years 5 mg; 15 years and over 10 mg; once daily in the evening
Pregnancy	Use if needed

2 CHAPTER TWO How it works

Mechanism and handling.

2.1 Mechanism

Cysteinyl leukotrienes (LTC4, LTD4, LTE4) are released by mast cells and eosinophils. In **aspirin-exacerbated respiratory disease**, inhibition of cyclo-oxygenase shunts arachidonic acid into the lipoxygenase pathway and floods the airways with leukotrienes, which is why leukotriene modifiers help these patients.

3 CHAPTER THREE Using it in ENT

Indications.

3.1 Allergic rhinitis

Less effective than nasal steroids and similar to oral antihistamines; most useful when asthma coexists.

3.2 Aspirin-exacerbated respiratory disease

Asthma, nasal polyps and respiratory reactions to aspirin and NSAIDs: montelukast helps nasal and asthma symptoms alongside nasal steroids, surgery and, in selected patients, aspirin desensitisation or biologics.

3.3 Childhood sleep-disordered breathing

Montelukast with a nasal steroid for 12 weeks reduces the apnoea-hypopnoea index in children with mild sleep apnoea and adenotonsillar hypertrophy; it is an option when surgery is not wanted or not yet possible.

4 CHAPTER FOUR Safety

What goes wrong.

4.1 Adverse effects

- **Neuropsychiatric reactions**: sleep disturbance, nightmares, agitation, depression and, rarely, suicidal thoughts. Regulators (MHRA, FDA) require warnings; tell families and stop the drug if they occur.
- Headache, abdominal pain; rarely eosinophilic granulomatosis with polyangiitis is unmasked as steroids are reduced.

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- The leukotriene shunt in **aspirin-exacerbated respiratory disease**.
- Its limited role compared with nasal steroids.
- The **neuropsychiatric** warning.

5.2 Common pitfalls

COMMON PITFALLS

- Starting montelukast in a child without warning the family about mood and sleep effects.

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